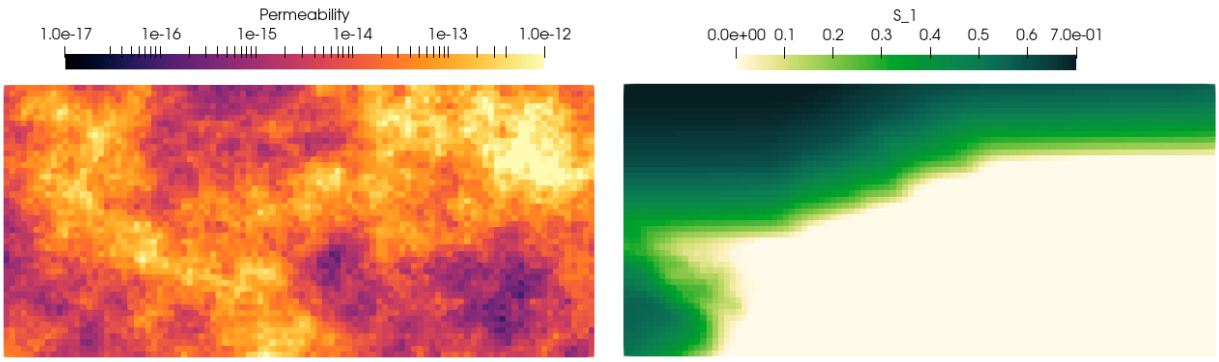
DARSim Example:

2D CCS with Heterogeneity (SPE10 Slice)



This example demonstrates how to use a heterogenous permeability field. In this case, the example uses a 2D slice of the SPE10 benchmark simulation permeability field, as shown above. The injection point is placed in the bottom left of the domain, and the production well is placed in the top right. After approximately 70 years of injection, the CO_2 saturation distribution is shown above.

To use the heterogenous field, the input file `Sim.txt` defines the permeability file to use (Perm_mSPE10T_v1.txt):

```
PERMEABILITY_UNIT
      mD

PERMEABILITY_SCALE
      Linear

      INCLUDE
      PERMEABILITY_X
            Perm_mSPE10T_v1.txt

      INCLUDE
      PERMEABILITY_Y
            Perm_mSPE10T_v1.txt

      INCLUDE
      PERMEABILITY_Z
            Perm_mSPE10T_v1.txt
```